Institute of Astronomy

Natural Sciences Tripos

Part III/MASt Astrophysics



Research Project Booklet

2025/26

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Part I: Key Information

Key Dates

Type	Milestone	When?		Format	Length	Where?
		Time	**Date**			
Submission	Interim Progress Report	12:00 noon	Last day of Full Michaelmas Term	PDF - LaTeX/equivalent	1,000 words	Moodle
Submission	Practice Presentation	12:00 noon	Last Monday of Full Lent Term	Microsoft PowerPoint	N/A	Moodle
Oral	Practice Oral Presentation	TBC	Last Tue. – Fri. of Full Lent Term	N/A	30 minutes	Hoyle Committee Room
Submission	Draft Final Report to Supervisor I	N/A	Last day of the Easter Vacation	PDF - LaTeX/equivalent	30 pages	Various
Submission	Final Report	12:00 noon	2nd Monday of Full Easter Term	PDF - LaTeX/equivalent	30 pages	Moodle
	Oral Presentation Slides			Microsoft PowerPoint	N/A	
Oral	Oral Presentation	TBC	2nd Tue. – Fri. of Full Easter Term	N/A	30 minutes	Hoyle Committee Room

The Part III/MASt Calendar

We **strongly advise** adding this calendar to your own, as it will be updated throughout the year with important events, deadlines and more.

- To add to an existing Google Calendar, click here.
- If you use another calendar, right click, copy and paste this link into your own calendar. Please note, simply clicking on the link will download a calendar file or import a copy directly into your calendar (depending on the software you use) but the information will not update as changes are made.

Format

The Written Report

Formal Requirements

- The Interim Report must not exceed 1,000 words in length
- The Final Report must not exceed 30 pages in length, including The Abstract, Figures, Tables, References, and Appendices (where applicable)
- The minimum acceptable font size is 11pt with at least single line spacing
- The text must be in single column format
- Margins must not be less than 2cm
- Figures must be legible when printed on A4 paper

Projects not meeting these requirements will be returned for revision and a penalty may apply for late submission at the discretion of the examiners.

Content

The Interim Progress Report

The Interim Progress Report, which may contain material that can be incorporated in the final project report, must indicate the progress made so far and show preliminary results. It should also give a clear indication of the project aims and a detailed plan of how these aims will be achieved. This is particularly important where the results of the project depend on data that has yet to be analysed.

There is no need for the Interim Progress Report to reiterate the material given in the Project Description.

The Final Report

The Final Report should read as a self-contained document, presented in the style of a scientific research report or paper in a scientific journal. The main sections of the report will describe the work undertaken, the results obtained and an assessment of their significance.

An Abstract, Introduction, Conclusions, References and appendices (where applicable) should also be included within the 30-page limit. Supporting Figures and Tables should be used both as an aid in presenting data and results and also to enhance the clarity of the submission. In some circumstances an appendix containing more extensive tabular material/results may be included.

The reader should be able to understand the context in which the investigation was undertaken, the main features of the project, the results and how they relate to the advancement of the subject. In

addition to the descriptive material, questions a report would be expected to address include, "Why were particular approaches adopted?" – back of the envelope calculations will often be helpful and relevant – "What has been learnt?" and "What information/work would have helped us to learn more?" You should take care to demonstrate that you have tested any analysis packages/codes that you use.

Structure

You will have read many research papers in the course of your project research. Some of these you will have found easy to read and others less so. You therefore already have a good idea of what makes a good write-up.

The Final Report should be logically structured, clear, and complete, while remaining concise. It is usual to set out the write-up in sections that include an introduction, a description of methods, results, and conclusions. The introduction should set out the problem to be solved, including why it is interesting, and previous work done. The methods section should describe what you have done in sufficient detail that the work can be reproduced by a reader. It is important to make clear what new work you have done yourself in this section. In the results section describe what you have found. Try to make it very clear which are the most interesting outcomes of the project. In the conclusion explain whether or not you have solved the problem you set out to solve. If so, explain how and if not then why not. You can also describe future work that might get closer to or verify your solution.

There are some points to particularly bear in mind:

1. Remember that your readers may not be experts in the field of your project. Begin your description from basic physical principles and describe how any observations have been made.
2. Write short sentences. Long and convoluted sentences, with numerous sub-clauses, are hard to read and often grammatically incorrect.
3. Use named references, such as (Eggleton, Fitchett and Tout 1989), in the text. This is the style generally used by astronomers. It is much easier to read than a number reference style that requires continual cross-referencing.
4. Be concise. Well-written reports do not need to fill the page limit.
5. Include a limited number of pertinent figures. A good figure can replace many words, but many similar figures can often be replaced by a few words. Ensure that axes are labelled properly, lines are sufficiently thick, that points and labels are in a large enough font and that the main details of the figure are explained in the caption. Avoid making figures too cluttered and do not include anything that is not relevant to your discussion.
6. Appendices are for additional reading only. The examiners will base their marking on the main report.

The Oral Presentation

Formal Requirements

- The Oral Presentation must consist of a 20-minute description of the project followed by up to 10 minutes of questions from the Part III/MASt Examiners.
- The written presentation must be submitted as a Microsoft PowerPoint (.pptx) file.
- If your Project's Supervisor, Co-Supervisor or University Teaching Officer has been appointed as the Senior Examiner or one of the Examiners, then to mitigate against any conflicts of interest, they will not ask you any questions during your oral presentation.

Referencing and Plagiarism

It is a requirement that the project investigation and the project report are both the work of the candidate alone and no form of collaboration is allowed.

It is a fundamental tenet of scientific research that due acknowledgment is given to the work and ideas of others that form the basis of, or are incorporated in, a research presentation. You must always acknowledge the source of an idea or material you use with a specific reference. Plagiarism, including the use of another individual's ideas, data, or text, is regarded as an extremely serious disciplinary offence by the University. For further information, please see the University's guidelines on Plagiarism and Academic Misconduct.

Guidelines for using Artificial Intelligence-powered tools

"A student using any unacknowledged content generated by artificial intelligence within a summative assessment as though it is their own work constitutes academic misconduct unless explicitly stated otherwise in the assessment brief." Plagiarism and Academic Misconduct

If you decide to use AI-powered tools for the report preparation you must state clearly which portions of the report were created with AI assistance. Note that even advanced versions of these tools can make mistakes and provide biased information. If you choose to use AI agents, it is your responsibility to verify the facts supplied and correct the logic underlying the arguments provided. Ultimately, you are being marked for your effort, so the AI-generated content can only be used to supplement your work not to substitute it. The core of the report must be based on your own ideas, your own analysis, your own conclusions and your own understanding of the topic. Finally, the above guidelines should not be taken as an endorsement or recommendation for use of these tools in your work.

Assessment

The research project element of the NST Part III Astrophysics course accounts for 6 assessment units (equivalent to the marks assigned to two 24-lecture Mathematics Part III lecture courses).

Distribution of Marks

The Interim Progress Report

The Interim Progress Report does not constitute part of the formal assessment but is regarded as an essential part of the monitoring procedure. The Course Coordinator will assess these reports and provide feedback to students and supervisors.

The Mock Oral Presentation

This practice presentation, delivered to an audience including the Part III/MASt Course Coordinator, Project Supervisors and UTOs, is not formally assessed but offers the opportunity to become familiar with the format of the presentation to be assessed by the Part III/MASt Examiners in the Easter Term. Students are strongly encouraged to attend the practice talks of their peers to further develop their understanding of and exposure to various presentation techniques.

The Final Draft Report

The Final Draft Report does not constitute part of the formal assessment but provides students with an opportunity for their work to be read by their Primary Project Supervisor, who will provide feedback during the final supervision.

The Final Report

The research project written report counts for 5 out of the total of 6 units of assessment assigned to the research project i.e., approximately 85% of the marks for the research project will be assigned on the basis of the written final report.

The Final Oral Presentation

The research project oral presentation counts for 1 out of the total of 6 units of assessment assigned to the research project i.e., approximately 15% of the marks for the research project will be assigned on the basis of the assessed oral presentation.

Examiners Criteria for Marking the Project Report and Oral Presentation

If your Project's Supervisor, Co-Supervisor or University Teaching Officer has been appointed as the Senior Examiner or one of the Examiners, then to mitigate against any conflicts of interest, they will not ask you any questions during your oral presentation, and they will also not mark your report.

The Examiners will award marks under three broad headings:

I. Scientific understanding

II. Quality of the research

III. Presentational and communication skills.

The format and timetable for submission form part of the Examination process. In their assessment of the project, the Examiners will take account of any breaches of the guidelines, including exceeding the word limit and late submission of the report.

Oral Presentation

The Examiners assessment will take into account the following:

- Visual Material: including relevance, clarity, attractiveness
- Oral Presentation: including overall structure, clarity, time keeping
- Response to Questions: including grasp of subject material, precision of answers

Written Project Report

- The Examiners will assess the report under the following headings:
- Overall structure and clarity of the report
- Planning, organisation, and prosecution of the research
- Understanding of the physics and the general scientific content
- Technical proficiency
- Analytical and Interpretational skills
- Significance of the results

Extension Requests and Illness

The Institute of Astronomy expects students to meet the advertised deadlines for the submission of all coursework, to ensure fairness to all students taking the course and allow prompt marking by the Department.

Your Part III Project makes up one third of total available Tripos marks and is subject to special arrangements.

The Final Report and Final Presentation Slides are linked with compulsory attendance at the Final Oral Presentation. As such, it is not possible for the Department to grant an extension of more than 48 hours. The following policy builds around point 7 of the University's Policy on managing extensions to Dissertations and coursework policy.

Extension Requests for the Final Report and/or Final Presentation Slides of up to 48 hours

Students may self-certify via this form for an extension of up to 48 hours from the original submission deadline.

Requests must be submitted **no later than 7 days before the original submission deadline** to allow for potential amendments to be made to the Final Oral Presentation schedule.

Extension Requests for the Final Report and/or Final Presentation Slides of over 48 hours

Late submission of over 48 hours is a particularly serious matter, and will require your College to apply to the Examination Access and Mitigation Committee (EAMC) to seek permission for any credit to be allowed. If this is not granted, the late work will not be counted.

The College **must** apply to the Secretary of the EAMC for an extension to a deadline and should **not** approach the Chair of Examiners, Course Coordinator or Senior Examiner directly.

Applications must be submitted **no later than 7 days before the original submission deadline** to enable the Secretary to consult the relevant Chair of Examiners or, where applicable, the Senior Examiner.

Penalties for late submission

You are required to self-certify prior to the original submission deadline (or at the point of submission at the very latest). Retrospective requests will not be accepted.

Where a student does not self-certify and does not submit their Final Report and Final Presentation Slides by the original submission deadline, zero marks will be awarded.

Where a student has self-certified and does not submit their Final Report and Final Presentation Slides by the revised submission deadline, zero marks will be awarded unless a further extension has been granted by the EAMC (see below).

Where a student has applied for a further extension via the EAMC and does not submit their work by the agreed submission date, zero marks will be awarded.

Illness during the Oral Presentations

The department defers to University Policy with regards to illness during an examination, including the Final Oral Presentation.

For full details, see the Policies and Guidance for Students here.